TREEWALK NYC • STEAM LESSON PLAN

Callery Pear

Pyrus calleryana

Species background: Native to China and Vietnam; introduced to the US in 1917 by plant explorer Frank Meyer as rootstock to help fight a disease affecting European pear orchards. Its synchronized, all-at-once white spring bloom -- often before any leaves appear -- makes it one of the most visually striking street trees for about a week each April.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

GRADE BAND K-2

Callery Pear: Identify callery pear by its shiny, teardrop-shaped leaf and springtime flower burst.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Look for a shiny leaf shaped like a teardrop. In spring, callery pears burst into white flowers seemingly overnight.

Procedure

1. Engage: Ask what a 'teardrop' shape looks like, then find it in the leaf.
2. Explore: If it's spring, count how many flowers you can see on one small branch.
3. Explain: Explain that this tree blooms all at once, almost overnight, which is why it stands out on the block.
4. Art: Draw a teardrop shape, then add dots all around it for flowers.
5. Evaluate: Student can describe the leaf shape and the springtime flower burst in their own words.

Discussion questions

Why might a tree bloom all at once instead of a few flowers at a time?

Assessment

Correctly identifies the teardrop leaf shape and describes the flower burst.

Extension

Look for this tree in a different season and compare what it looks like without flowers.

GRADE BAND 3-5

Callery Pear: Explain mass flowering and observe pollinator activity around this tree.

Standards connection: NGSS 3-LS4-3; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

In spring, callery pears burst into white flowers all at once -- that attracts every nearby pollinator on the same days, a strategy called mass flowering.

Procedure

1. Engage: Ask why blooming on the exact same days as every other callery pear nearby might help this tree.
2. Explore: If possible, observe (or research) what pollinators visit the tree during its bloom week.
3. Explain: Introduce mass flowering as a pollinator-attraction strategy.
4. Art: In spring, sketch the tree covered in flowers using only small dots -- hundreds of white dots for hundreds of small blossoms.
5. Evaluate: Student writes two sentences explaining mass flowering in their own words.

Discussion questions

What might happen to pollination success if only one tree bloomed while all its neighbors bloomed a week later?

Assessment

Two accurate sentences describing mass flowering as a pollinator strategy.

Extension

Research one other mass-flowering plant and compare its bloom strategy.

GRADE BAND 6-8

Callery Pear: Explain how cultivar cross-pollination turned a popular street tree into an invasive species.

Standards connection: NGSS MS-LS4-4; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, paint (white and cream), fine brushes or cotton swabs.

Teacher background

Different callery pear cultivars can cross-pollinate into fertile, spreading offspring, which is why this once-popular street tree is now considered invasive in parts of the US.

Procedure

1. Engage: If the popular 'Bradford' cultivar was bred to be sterile, how could this species still spread and become invasive?
2. Explore: Research how different cultivars of the same species can cross-pollinate even when each individual cultivar can't self-pollinate.
3. Explain: Connect cross-cultivar pollination to the species' current invasive status in the eastern US.
4. Art: Create a pointillist-style painting of the tree in bloom using only dabs of white and cream paint -- no lines.
5. Evaluate: Write a paragraph explaining how a 'sterile' tree could still become an invasive problem.

Discussion questions

What responsibility do nurseries and cities have when they widely plant a single ornamental cultivar?

Assessment

Paragraph correctly explains cross-cultivar pollination as the mechanism of invasiveness.

Extension

Research a native tree that could responsibly replace callery pear as a street tree.

GRADE BAND 9-12

Callery Pear: Apply pointillist color theory and evaluate callery pear as a case study in unintended ecological consequences.

Standards connection: NGSS HS-LS4-2; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, paint, fine brushes.

Teacher background

Callery pear cultivars were bred to be individually sterile, but different cultivars planted near each other cross-pollinate into fertile, weedy offspring -- a case study in unintended consequences in ornamental horticulture.

Procedure

1. Engage: What does this case study reveal about the limits of predicting ecological outcomes from horticultural breeding decisions?
2. Explore: Research the timeline from callery pear's introduction as disease-resistant rootstock to its current invasive status.
3. Explain: Discuss why breeding for one trait (sterility, disease resistance) doesn't guarantee ecological safety.
4. Art: Research Georges Seurat's pointillism technique, then apply it to a study of this tree's mass-flowering display, explaining with color theory how dots of paint blend optically at a distance.
5. Evaluate: Submit the painting plus a short written case-study analysis proposing a native replacement tree.

Discussion questions

What lessons from callery pear's history should inform how new ornamental cultivars are evaluated before wide release?

Assessment

Rubric covering historical accuracy, ecological reasoning, and the painting's technical use of color theory.

Extension

Compare callery pear's invasive mechanism to Norway maple's -- are the ecological risks similar or different?

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.